

M-Pesa: Simple Tech, Massive Impact

How a Backend-First Solution Built
Africa's Financial AP

Presented by [Karobia Backend-Focused Software
Engineer]



Financial Exclusion

The World Before M-Pesa

1. Banking was for the urban elite (less than 20% population).
2. Sending money required travel: costly, slow, and risky.
3. Cash economy limited savings, loans, and business growth.



USSD & SMS Technology

Accessible financial services through simple, secure mobile technology for all Kenyans

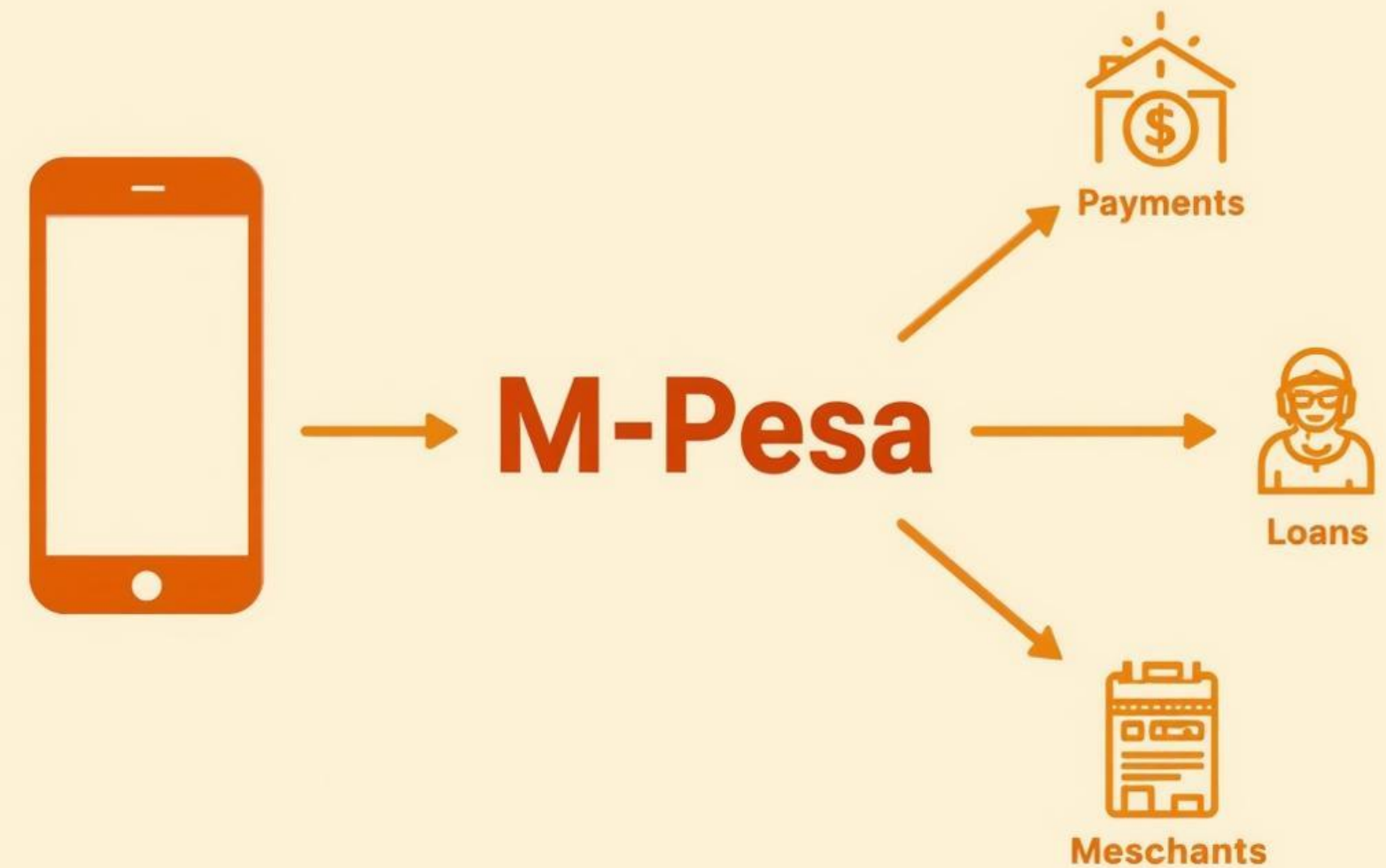
- Works on any mobile phone, **no internet needed**
- **Secure, simple, fast** transactions for users
- Bridges the gap between people and **financial services**



Simple, Powerful, Scalable

M-Pesa's architecture enables seamless integration without user complexity or confusion

- Just like an API: standardized inputs, reliable outputs
- Integrates multiple services effortlessly
- Users remain unaware of backend complexities

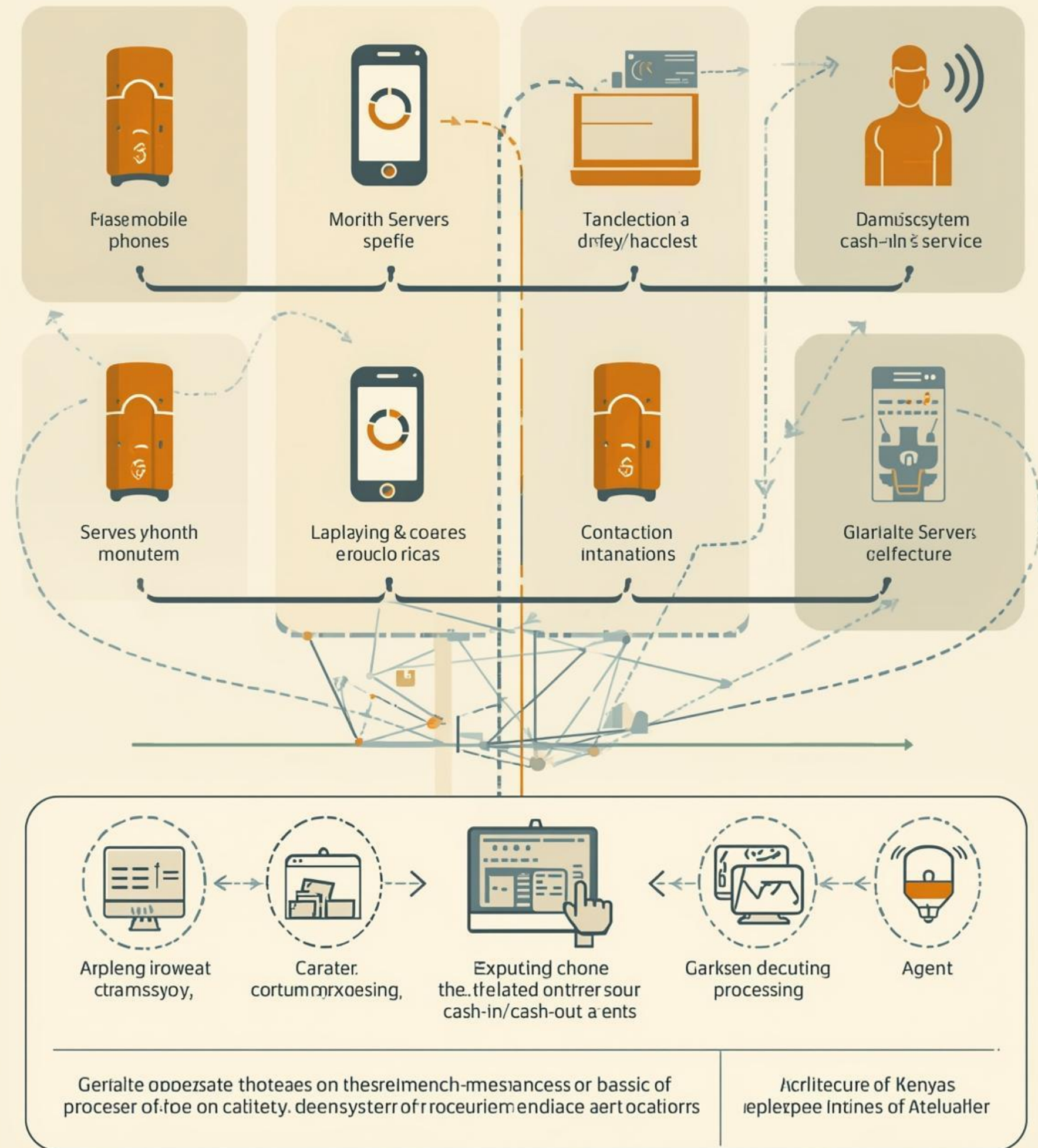


Backend & Agent Network

Transaction processing, physical agents, and a robust financial network in Kenya

- Backend: transaction processing, security, scalability
- Agents: physical presence for cash-in/cash-out
- Reliable network ensures system robustness

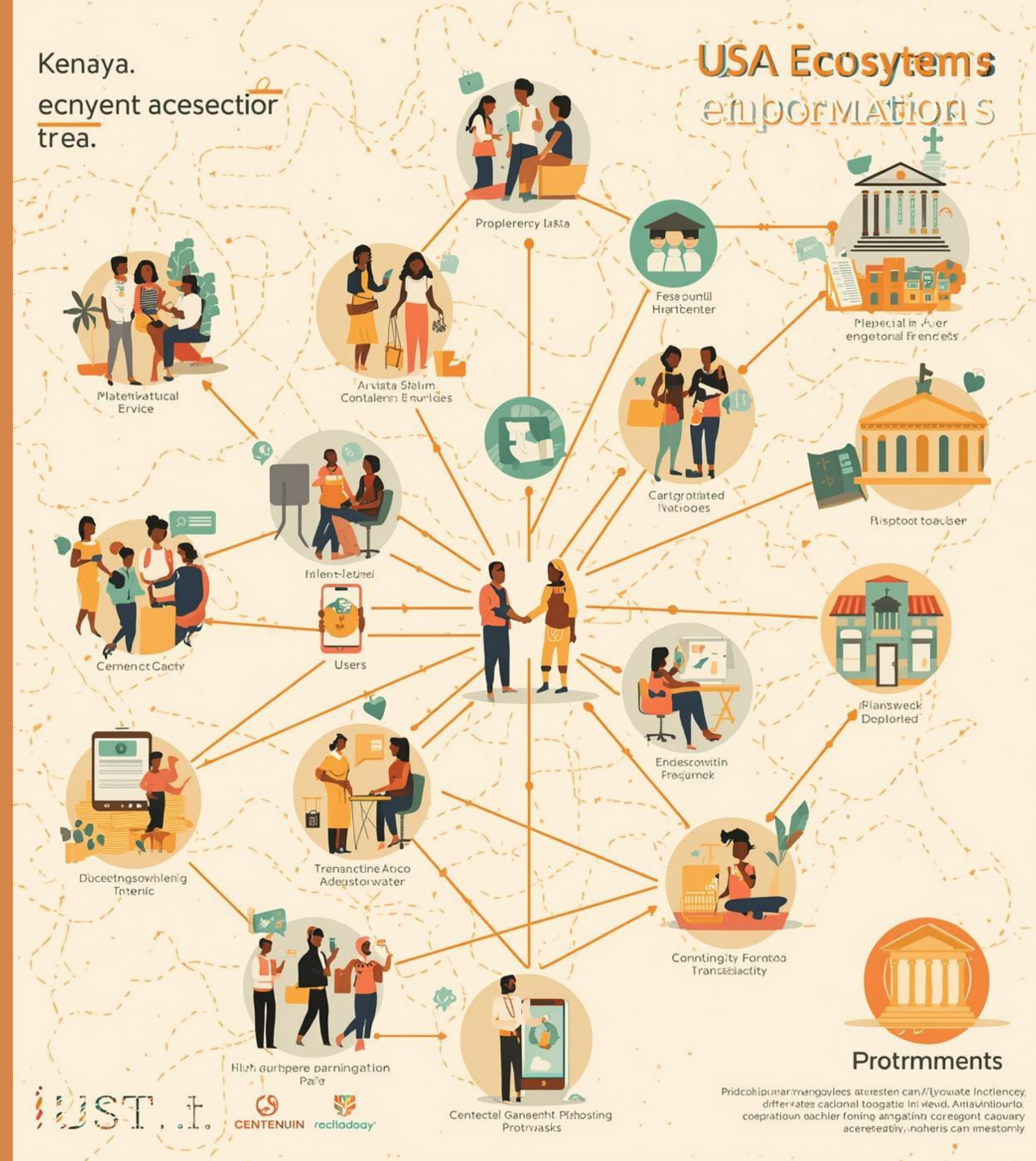
M-PeA System ARchitectiuns?



75%

Ecosystem Connectivity

75% of Kenyans now have access to financial services through M-Pesa, significantly bridging the gap between users, merchants, banks, and government services, enhancing economic participation.



Lessons from M-Pesa

Key engineering insights driving innovation and impact in financial technology solutions

- Robustness – transactions must never fail
- Context-awareness – adapt to mobile limitations & rural settings
- Platform-thinking – enable third-party services & innovation





Transforming Access

Simple technology can drive **massive social change** in communities

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